

CORE PURE / FURTHER CALCULUS

Integration by Substitution

Questions by topic

15 questions · 113 marks

Questions

Question 1	8 marks	2
Question 2	8 marks	3
Question 3	6 marks	4
Question 4	8 marks	5
Question 5	8 marks	6
Question 6	9 marks	7
Question 7	6 marks	8
Question 8	7 marks	9
Question 9	8 marks	10
Question 10	7 marks	11
Question 11	7 marks	12
Question 12	7 marks	13
Question 13	8 marks	14
Question 14	7 marks	15
Question 15	9 marks	16

1. Show that

$$\int_0^{\pi/2} \frac{\cos^5 \theta}{2 + \sin \theta} d\theta = 9 \ln\left(\frac{3}{2}\right) - \frac{41}{12} \quad [8]$$

2. (a) Use the substitution $u = 1 + x^2$ to show that

$$\int_0^2 \frac{x^9}{1+x^2} dx = \int_p^q \frac{(u-1)^4}{2u} du$$

where p and q are constants to be found.

[3]

- (b) Hence show that

$$\int_0^2 \frac{x^9}{1+x^2} dx = A + B \ln 5$$

where A and B are constants to be found.

[5]

3. Show that

$$\int \frac{\sqrt{x^2 - 16}}{x^2} dx = \operatorname{arcosh}\left(\frac{x}{4}\right) - \frac{\sqrt{x^2 - 16}}{x} + c \quad [6]$$

4. Find the exact value of

$$\int_0^{\frac{1}{2} \ln(2+\sqrt{3})} \frac{\operatorname{sech}^2 x}{\sqrt{4-9 \tanh^2 x}} dx$$

[8]

5. Using calculus, find the exact values of

(i)

$$\int_1^2 \frac{1}{x^2 + 3x + 2} \, dx \quad [3]$$

(ii)

$$\int_1^{\sqrt{3}} \frac{\sqrt{4 - x^2}}{x^2} \, dx \quad [5]$$

6. Find the exact value of

$$\int_2^4 \frac{\sqrt{x^2 - 4}}{x^2} dx$$

Give your answer in simplest exact form.

[9]

7. Show that

$$\int \frac{x^2}{\sqrt{x^2 + 16}} \, dx = \frac{1}{2}x\sqrt{x^2 + 16} - 8 \operatorname{arsinh}\left(\frac{x}{4}\right) + c \quad [6]$$

8. Show that

$$\int_0^{\ln 2} \frac{e^{3x} + 2e^{2x}}{\sqrt{e^x + 4}} \, dx = \frac{32\sqrt{6} - 30\sqrt{5}}{5} \quad [7]$$

9. Using the substitution $u = \frac{2}{3} \tanh x$, evaluate

$$\int_0^1 \frac{\operatorname{sech}^2 x}{\sqrt{9 - 4 \tanh^2 x}} \, dx \quad [8]$$

10. Show that

$$\int_0^{\pi/2} \frac{\sin 3\theta}{2 + \cos \theta} d\theta = 15 \ln \left(\frac{3}{2} \right) - 6 \quad [7]$$

11. (a) Use the substitution $u = \sqrt{2x+1}$ to show that

$$\int_0^{15/2} \frac{x}{2 + \sqrt{2x+1}} dx = \int_p^q \frac{u(u^2 - 1)}{2(u+2)} du$$

where p and q are constants to be found.

[3]

- (b) Hence show that

$$\int_0^{15/2} \frac{x}{2 + \sqrt{2x+1}} dx = A - B \ln 2$$

where A and B are constants to be found.

[4]

12. Use an appropriate substitution to find the exact value of

$$\int_0^{\pi/6} \frac{\cos x \cos 2x}{(1 + \sin x)^{3/2}} dx$$

Give your answer in the form $a + b\sqrt{6}$, where a and b are rational numbers to be determined.

[7]

13. Using calculus, find the exact values of

(i)

$$\int_{-2}^0 \frac{1}{x^2 + 4x + 8} \, dx \quad [3]$$

(ii)

$$\int_0^{\sqrt{3}} \frac{x^2}{\sqrt{4 - x^2}} \, dx \quad [5]$$

14. Show that

$$\int_1^3 \frac{(x-1)^2}{\sqrt{x+1}} \, dx = \frac{16}{15}(7-4\sqrt{2}) \quad [7]$$

15. Use an appropriate substitution to find the exact value of

$$\int_0^{2\sqrt{2}} \frac{x^2}{\sqrt{16-x^2}} \, dx$$

Give your answer in the form $a + b\pi$, where a and b are rational numbers.

[9]